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Laborator 2 EP

Task 1

Generational garbage collection - obiectele sunt impartite sau partitionate in generatii multiple. Fiecare obiect nou intra in prima generatie, daca un obiect supravietuieste unei runde de garbage collection, trece intr-o generatie mai mare. Generatiile mai mici sunt colectate mai des.

Deoarece cele mai multe dintre obiectele noi create mor repede, mareste performanta garbage collector-ului si reduce timpul de pauza al acestuia. Pentru a decide cand ruleaza, fiecare generatie are un counter si o limita individuala. Counterul retine numarul de alocari minus dealocari de la ultima colectie. De fiecare data cand aloci un nou obiect, CPython verifica daca counterul din prima generatie e mai mare decat valoarea limitei sale. Daca da, Pythonul initializeaza procesul de colectare.

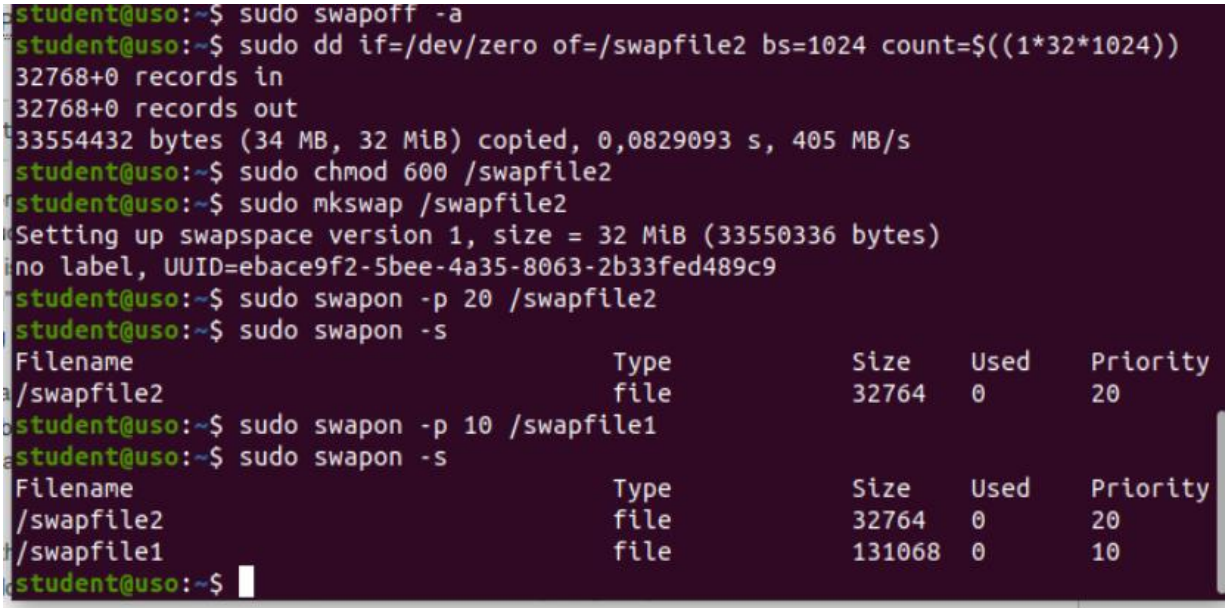
```
student@uso:~$ python3
Python 3.8.2 (default, Jul 16 2020, 14:00:26)
[GCC 9.3.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> import gc
>>> gc.get_threshold()
(700, 10, 10)
>>> gc.get_count()
(508, 9, 0)
```

Task 2

Primul swapfile

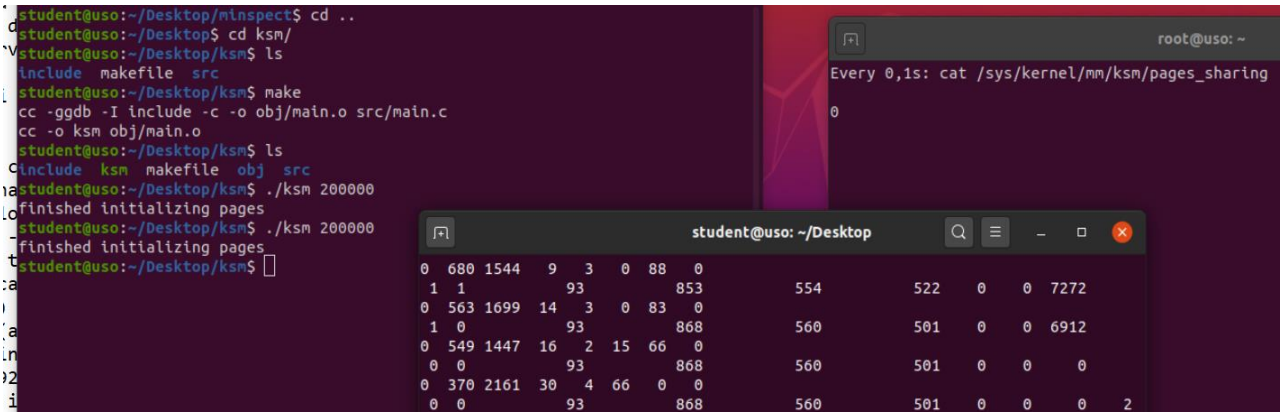
```
student@uso:~$ sudo swapoff -a
[sudo] password for student:
student@uso:~$ sudo dd if=/dev/zero of=/swapfile1 bs=1024 count=$((1*128*1024))
131072+0 records in
131072+0 records out
134217728 bytes (134 MB, 128 MiB) copied, 0,43079 s, 312 MB/s
student@uso:~$ sudo chmod 600 /swapfile1
student@uso:~$ sudo mkswap /swapfile1
Setting up swapspace version 1, size = 128 MiB (134213632 bytes)
no label, UUID=31804173-54ef-4679-a612-9648622f03e4
student@uso:~$ sudo swapon -p 10 /swapfile
student@uso:~$ sudo swapon -p 10 /swapfile1
student@uso:~$ sudo swapon -u /swapfile1
swapon: invalid option -- 'u'
Try 'swapon --help' for more information.
student@uso:~$ sudo swapon -s /swapfile1
Filename                                Type            Size    Used    Priority
/swapfile                               file            752960  0       10
/swapfile1                              file            131068  0       10
```

Al doilea swapfile

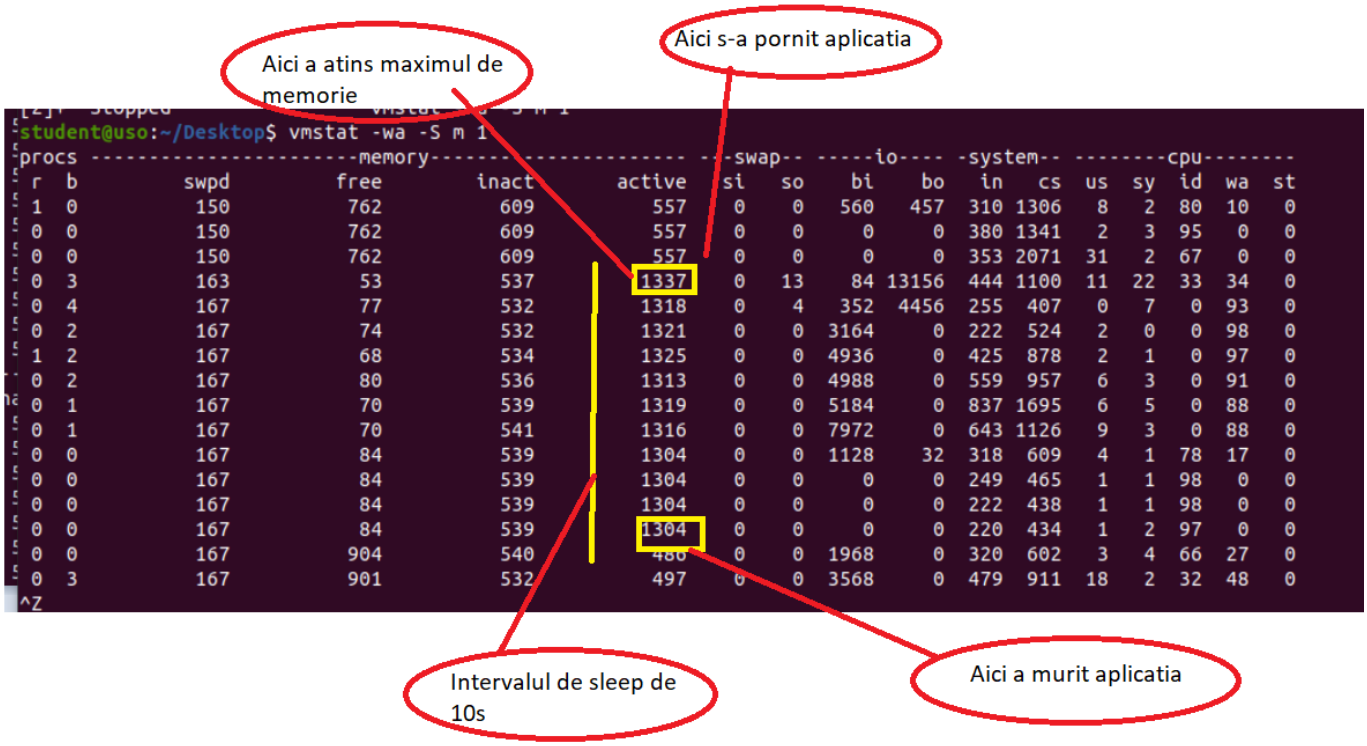


Task 3

Am modificat valorile in pages_to_scan si max_page sharing la 1000, respectiv 1 milion, am rulat programul:



Am editat imaginea cu terminalul cu comanda vmstat pt a prezenta datele cerute, dupa cum urmeaza:



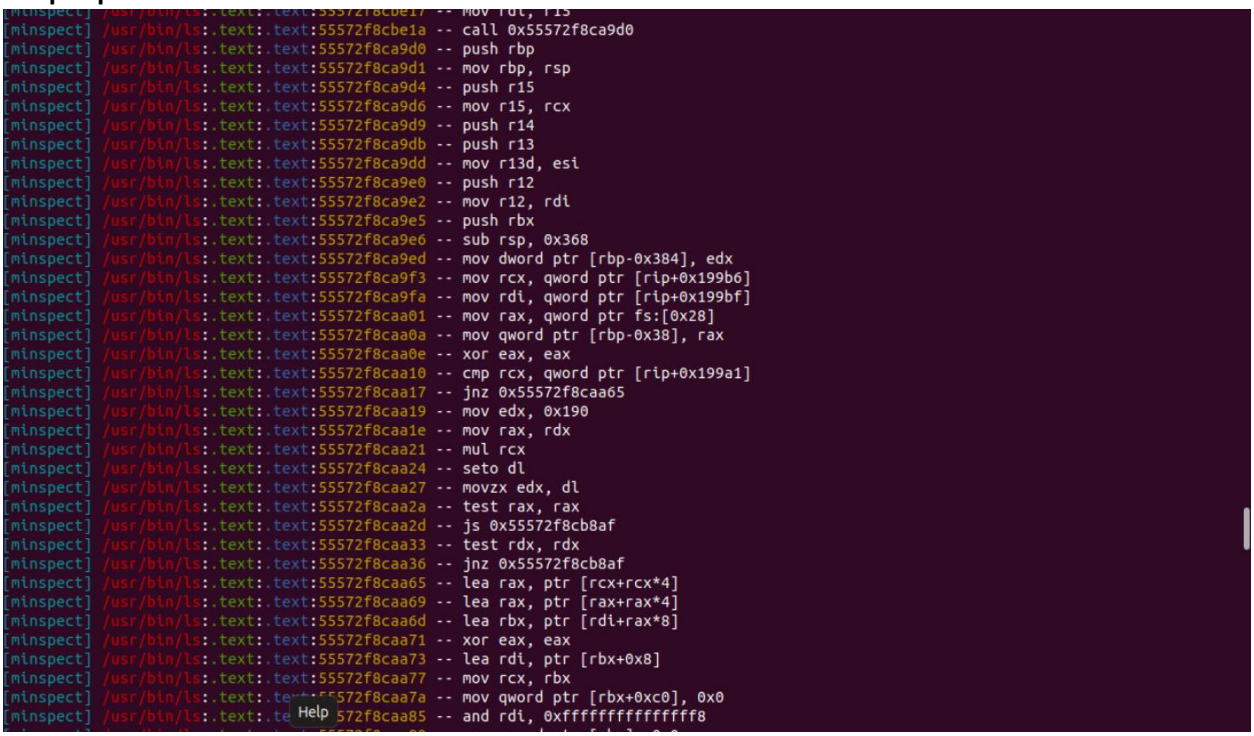
Deci programul incepe fix inainte sa creasca brusc numarul de pagini active (intre momentul in care sunt 557 pagini active si cel in care sunt 1337), atinge maximul cand sunt atinse nr maxim de pagini active (in cazul nostru imediat dupa pornire), are apoi intervalul de sleep de 10s, iar apoi moare cand nr de pagini active revine la valoarea aproximativ de dinainte de a se porni programul.

Task 4

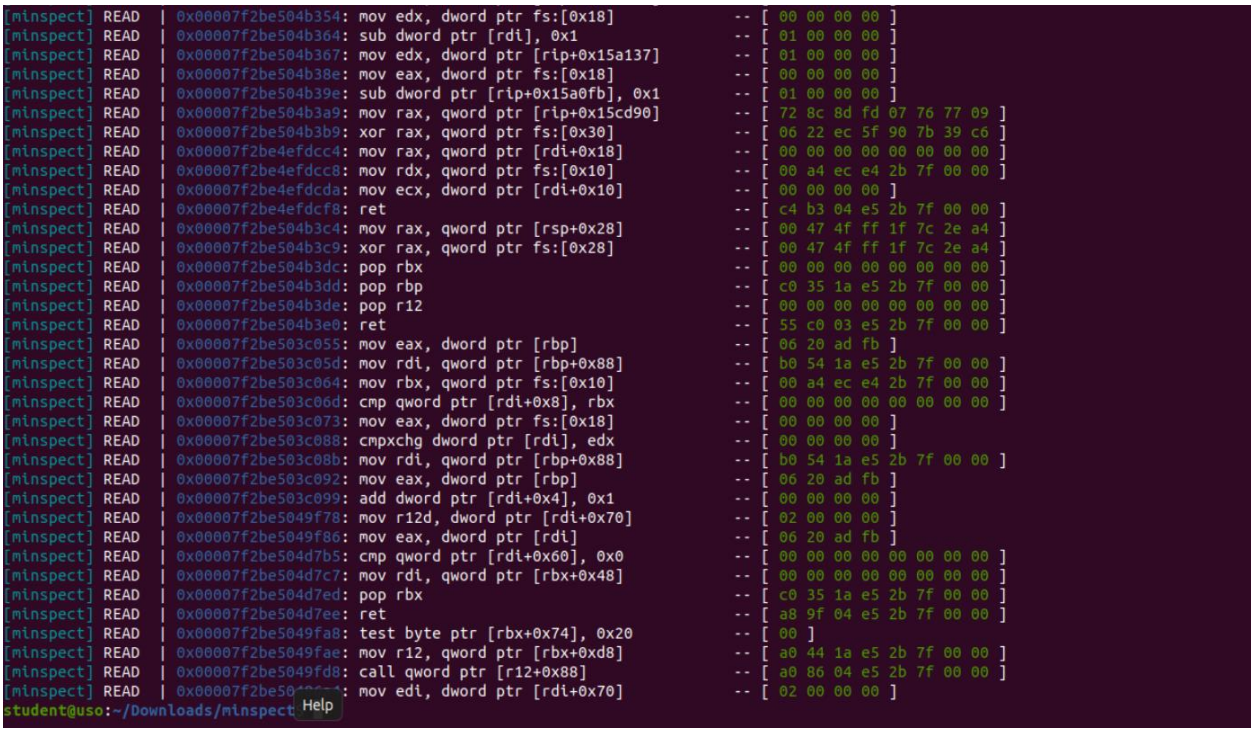
Comanda care se aplica pt patchurile A, B, C:

```
ime/pincrt/crtendS.o -lpin3dwarf  -ldl-dynamic -nostdlib -lstlport-dynamic -lm-dynamic -lc-dynamic -lunwind-dynam
make[1]: Leaving directory '/home/student/Downloads/minspect'
student@uso:~/Downloads/minspect$ ./third_party/pin-3.13/pin -t obj-intel64/minspect.so -- /bin/ls 1>/dev/null
```

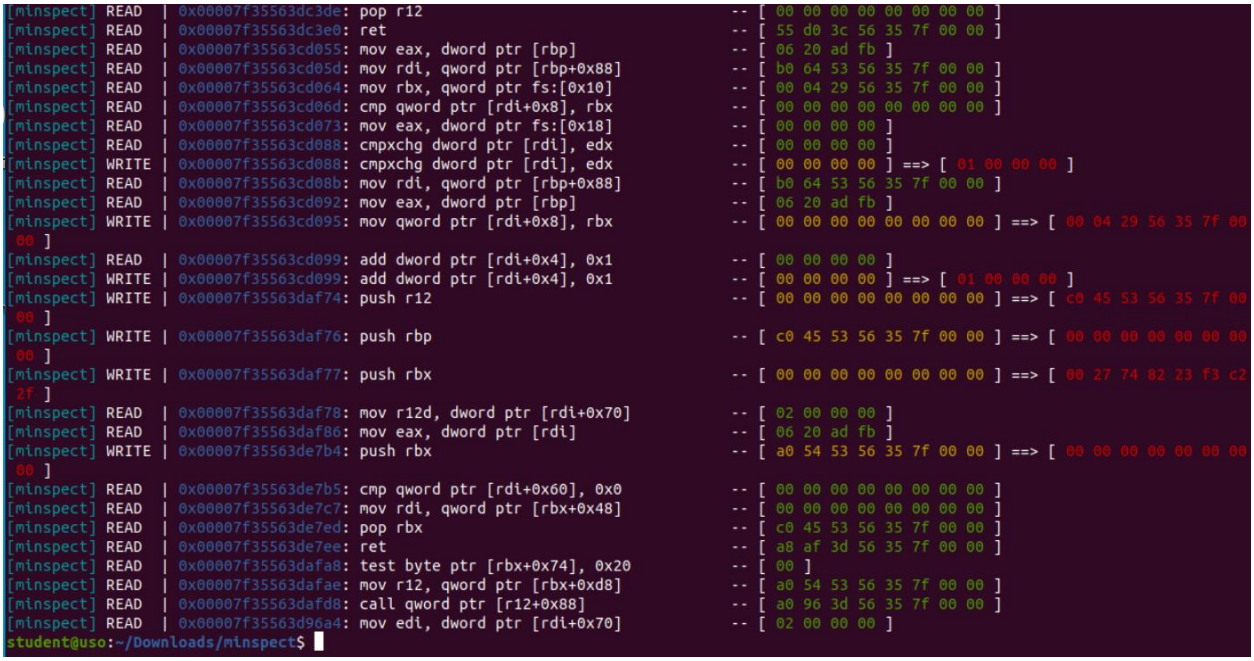
Output patchul A:



Output Patchul B:



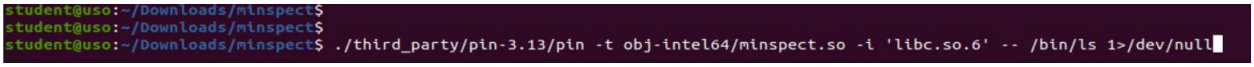
Output patchul C:

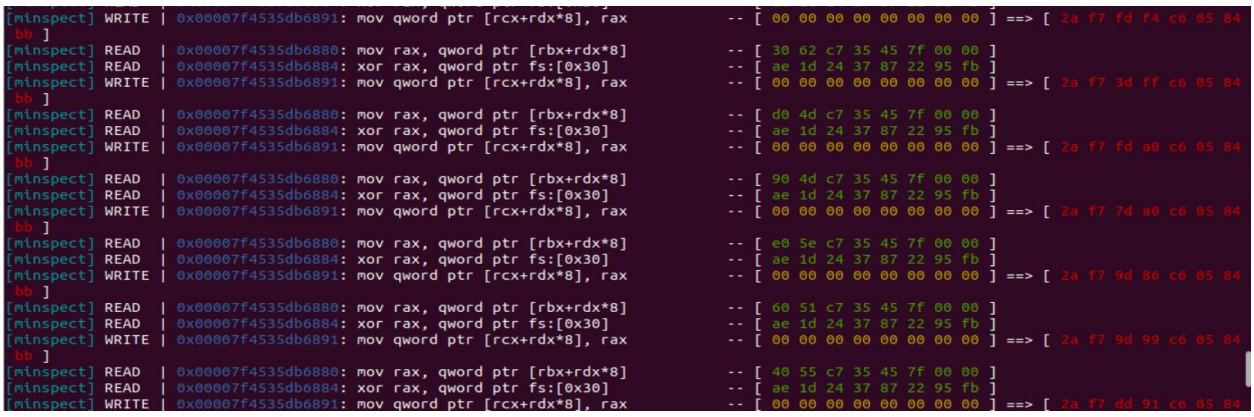


Putem vedea cum se afiseaza diferite detalii, de exemplu la patch-ul C, pentru scrieri se va afisa starea memoriei inainte si dupa ce s-a afisat o chestie, pentru a vedea cum s-a modificat. Desi se insereaza functia de analiza si inainte si dupa scriere, cea dinainte salveaza starea memoriei intr-o stiva, si cea de dupa afiseaza diferenta.

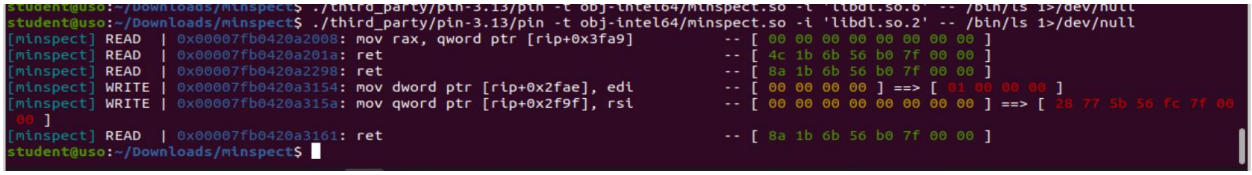
Comanda si output patchul D:

Pt libc.so.6 (binarul in sine)



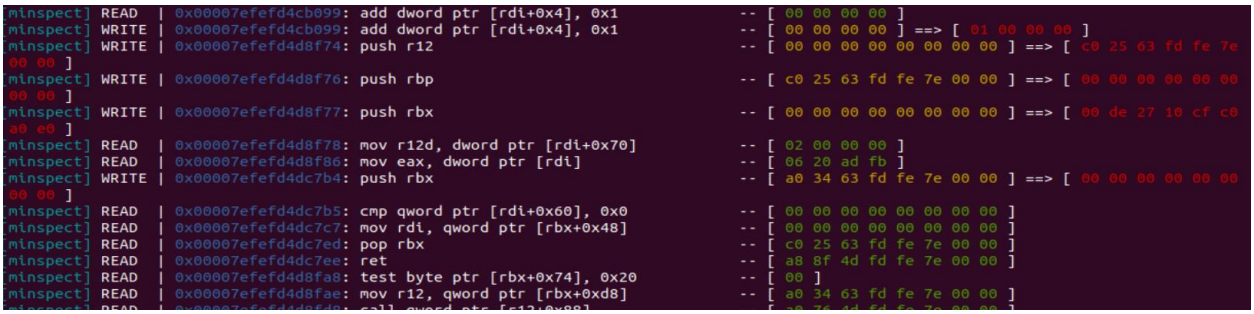


Pt libdl.so.2



Se pot da si comenzi de tipul acesta, cu mai multe tipuri:





Task 5

